

MTH209: Assignment 10

Minimizing a non-convex function

In this assignment, we want to minimize a non-convex function $f(x)$ that has a unique global minima, x^* . That is, we want to find

$$x^* = \arg \min_{x \in \mathbb{R}} f(x).$$

The problem is that any numerical scheme will typically converge to a local minima, which may or may not be a global minima. Further, the `optim` function, which we use with Nelder-Mead option, may not always converge to a minima.

In order to tackle this, I have written a function, `find_minima` that takes a function that is compatible with `optim`, runs `optim` for various starting values, and returns the best guess of x^* . My code below contains errors: logical and syntactical, that you need to fix.

```
# fun: function, compatible with optim
# start: a vector of starting values
find_minima <- function(fun, start)
{
  n.values <- size(start)

  local.minimas <- numeric(length = n)
  fn.at.minimas <- numeric(length = n)

  for(i in 1:n)
  {
    fit <- optim(part = start, function = fun)

    local.minimas <- fit$value
    fn.at.minimas <- fit$par

    # most suitable functional values
    ind <- which.max(fn.at.minimas)

    # return best argument
    return(local.minimas[ind])
  }
}
```

```
}  
}
```

Below is a test function you can use to test (DO NOT SUBMIT THIS FUNCTION):

```
fx <- function(x)  
{  
  fun <- 1/(1 + (5 - x)^2) * 1/(1 + (2 - x)^2/.3)  
  return(-fun)  
}
```

NOTE: Only upload the function `find_minima` in `assignment10.R` and nothing else.